KARAN S. NAIR

Graduate Applicant | M.Tech in Data Science and Artificial Intelligence

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Computer science graduate with strong mathematical foundations and hands-on experience in Python, SQL and reproducible data analysis. Seeking postgraduate study to deepen understanding of statistical learning, optimisation and responsible AI through research and applied projects.

EDUCATION

B.E. in Computer Science and Engineering | Sample Institute of Technology (fictional)

2021-2025 | CGPA: 8.6/10

Coursework: Linear Algebra, Probability and Statistics, Calculus, Numerical Methods, Data Structures and Algorithms, Database Systems, and Machine Learning.

TECHNICAL AND MATHEMATICAL FOUNDATIONS

Programming and tools: Python, NumPy, pandas, scikit-learn, Matplotlib, SQL, PostgreSQL, SQLite, Git, Jupyter, Linux and pytest.

Mathematics: Vectors, matrices, dot products, derivatives, gradient descent, conditional probability, sampling, descriptive statistics and confidence intervals.

Data and evaluation: Missing-value handling, outlier analysis, aggregation, algorithmic complexity, train/validation/test separation, cross-validation, data leakage, regression metrics and reproducible experiments.

SELECTED PROJECTS

Household Electricity Demand Forecasting | Academic Capstone

- Prepared approximately 8,000 hourly observations from a public household-energy dataset. Checked missing readings, duplicate timestamps and chronological ordering before creating features.
- Compared a seasonal-naive baseline, linear regression and a random forest using time-ordered validation. Fitted preprocessing only on training data and evaluated MAE and RMSE on a held-out future period.
- Obtained a test MAE of 0.21 kWh against a baseline of 0.29 kWh. Personally implemented data cleaning, features, evaluation and the project report; documented that results from one household may not generalise to others.

Commuter Survey Analysis | Database and Statistics Coursework

- Analysed 500 synthetic commuter records using SQL joins, GROUP BY queries and pandas. Validated missing entries and inconsistent units; compared mean, median and variability of journey duration.
- Estimated uncertainty with bootstrap confidence intervals and distinguished observed associations from causal claims. Shared a data dictionary, seeded notebook and reproducible results through Git.

Gradient Descent from First Principles | Independent Study

- Implemented linear regression and mean-squared-error gradients in NumPy. Checked derivatives using finite differences and compared the fitted model with scikit-learn.
- Investigated how feature scaling and learning rate affect convergence. Added tests for empty inputs, invalid shapes and simple known solutions, and explained the findings in a technical notebook.

RESEARCH INTERESTS AND LEARNING READINESS

Interests: Reliable machine learning, data quality and energy analytics. Practices include documenting assumptions, crediting sources, reporting limitations and protecting personal data. Academic experience has involved collaborative reviews and presenting methods, results and improvements.

FICTIONAL CANDIDATE - FOR APPLICATION TESTING ONLY. All qualifications and results are invented.